

# BUS713 Financial Analytics: Workshop 5

## Derivatives: Futures and Options for Risk Management

*No R this week: focus on the numbers and what they mean for a business decision*

Modules 1 to 4 measured return, risk, and required return using historical data. Module 5 shifts to instruments firms use to manage risk before it happens: futures and options. Every task below is a business decision wearing a numerical problem. Work through each calculation, then answer the interpretation question that follows it; the number is never the point on its own. Work through each task in order.

### Task 1: Why Futures Exist and Why Contracts Are Standardised

**Scenario:** *A friend who trades stocks casually tells you futures contracts are "just a way to gamble on commodity prices." You disagree, and you have a concrete example to prove your point.*

A futures contract exists because two parties on opposite sides of the same price risk want certainty. Standardising every term except price is what makes the contract tradable by anyone, not just the two original counterparties.

1. A cocoa futures contract fixes contract size, quality grade, delivery month, and delivery location, leaving only price to be negotiated. Explain why fixing all of those terms in advance, rather than letting each pair of traders negotiate their own custom contract, is what gives futures markets their liquidity.
2. A chocolate manufacturer uses 50 tonnes of cocoa a month and is worried prices will spike before its next purchase. Explain, in business terms, what buying a cocoa futures contract today would let the manufacturer do, and why the goal is price protection rather than profit.
3. Respond to your friend directly: using the chocolate manufacturer example, explain why "hedging" with futures is economically different from speculating on commodity prices, even though both use the same contract.

### Task 2: Margin, Marking-to-Market, and the Clearinghouse

**Scenario:** *You're a risk manager training a new trader who has never posted margin before and is nervous about what happens if a position moves against them overnight.*

Initial margin is the deposit required to open a position. Maintenance margin is the minimum balance the account must stay above. A margin call restores the account back up to the initial margin level, not just to the maintenance level.

4. A copper futures contract is worth \$150,000 and requires an initial margin of 8%. Calculate the initial margin required in dollars.
5. Maintenance margin on this contract is \$10,500. After two days of adverse price moves, the account balance falls to \$9,200. Calculate the margin call amount required to restore the account back to the initial margin level from Question 1.
6. Explain, in your own words, why the clearinghouse requires daily marking-to-market rather than settling all gains and losses only when the contract expires months later. What risk does this eliminate that would otherwise build up over the life of the contract?

### Task 3: Hedging with Futures — Certainty Has a Cost

**Scenario:** *A coffee exporter's board wants to lock in revenue for a shipment due in six months, but one board member is worried about "missing out" if prices rise.*

7. The exporter plans to sell 5,000 bags of coffee in six months and sells futures today locking in \$180 per bag. Calculate the target revenue this hedge is designed to guarantee.
8. At delivery, the spot price has fallen to \$150 per bag. Calculate the physical sale revenue, the gain or loss on the futures position, and the net revenue. Did the hedge achieve the target from Question 1?
9. Now suppose instead the spot price at delivery had risen to \$210 per bag. Calculate the physical sale revenue, the gain or loss on the futures position, and the net revenue. In 2 to 3 sentences, explain to the worried board member why the company did not get to keep the extra profit from the higher price, even though the hedge worked exactly as designed.

### Task 4: Options Payoffs — Calls and Puts

**Scenario:** *Two investors are looking at the same stock and the same strike price, but one expects the price to rise and the other expects it to fall. You need to show each of them what they actually stand to gain or lose.*

A call option's payoff at expiration is the stock price minus the strike price, or zero if that is negative. A put option's payoff is the strike price minus the stock price, or zero if that is negative. Profit subtracts the premium paid from the payoff.

10. An investor buys one call option contract (100 shares) with a strike price of \$80 and a premium of \$4 per share. Calculate the investor's total profit or loss if the stock is trading at \$70, \$80, and \$95 at expiration.
11. A second investor buys a put option on the same stock, with the same strike (\$80) and the same premium (\$4 per share). Calculate that investor's total profit or loss at the same three prices: \$70, \$80, and \$95.
12. At what stock price does each option (the call and the put) exactly break even, ignoring transaction costs? Explain why the maximum loss for an option buyer is always capped at the premium paid, no matter how far the price moves against them.

### Task 5: Protective Puts vs Futures — Comparing the Hedge

**Scenario:** A soybean farmer is deciding between a futures hedge and a protective put for the upcoming harvest, and wants to see the actual dollar trade-off, not just the theory.

13. The farmer expects to harvest 8,000 tonnes in four months. Current spot price is \$500/tonne. Instead of futures, the farmer buys put options with a strike of \$500/tonne and a premium of \$20/tonne. Calculate the farmer's net revenue if the price at harvest falls to \$420/tonne.
14. Calculate the farmer's net revenue under the same put strategy if the price instead rises to \$560/tonne.
15. A futures hedge locked in at \$500/tonne would have produced \$4,000,000 of net revenue in both scenarios (use Task 3's logic to confirm this yourself). Compare that to your answers in Questions 1 and 2. In which scenario does the put strategy outperform the futures hedge, and by how much? What is the farmer giving up in the other scenario in exchange for that flexibility?

### Task 6: Choosing the Right Instrument — A Business Recommendation

**Scenario:** A CFO has asked the finance team to recommend futures or options for two different hedging situations, and then wants a final one-page recommendation for a third.

16. A mid-sized manufacturer imports a key raw material and has very tight cash reserves, with limited ability to post additional margin on short notice if prices move against it. Would you recommend futures or options for hedging this exposure? Justify your answer using the cash flow and margin-call characteristics of each instrument.
17. A different company wants full, guaranteed cost certainty for board-level budgeting purposes and is willing to give up any potential upside to get it. Which instrument fits this mandate, and why does the alternative not fully satisfy it?
18. Write a 4 to 6 sentence recommendation memo to a CFO who is undecided between the two instruments for hedging a \$10 million exposure, referencing at least one numerical concept from this workshop (margin, premium, payoff symmetry, or basis risk) to support your recommendation.

### Wrap-Up

You worked through why futures markets exist and what standardisation buys traders, calculated margin requirements and margin calls, hedged a physical position with futures and measured the opportunity cost of doing so, computed call and put payoffs from first principles, and compared a protective put to a futures hedge across two price scenarios. The recurring theme: every derivative reshapes a payoff, and every reshaping has a price.